Modeling Narrative Story Similarity via Representation Learning

Team #23

Moritz Gilges, Juri Hayashi, Seyed-Arsam Islami, Mohamed Abouelella

Problem/Task:

- Modeling narrative similarity between stories
- Narrative similarity dimensions: Abstract Theme, Course of Action, Outcomes.
- Writing style, lexical overlaps and named entities should not be considered
- Two complementary tasks:
  - **Track A** (Pairwise Similarity Judgement):
    Given an anchor story and two candidates, the system selects the one more narratively similar to the anchor
  - **Track B** (Representation Learning):
    Generate vector representations (embeddings) for each story, such that cosine similarity reflects narrative similarity

Data:

- Four different dataset are provided, containing in total 2139 labeled and 400 unlabeled triples
- Stories show recurrent narrative themes and structures
- High frequency of abstract, plot-level concepts

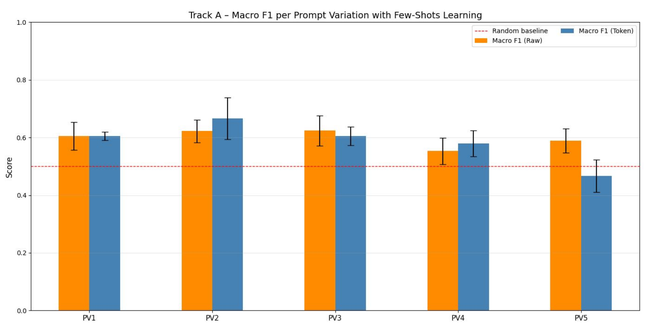
- Repeated entities may act as shortcut signals
- Potential bias toward common narrative motifs

Track A:

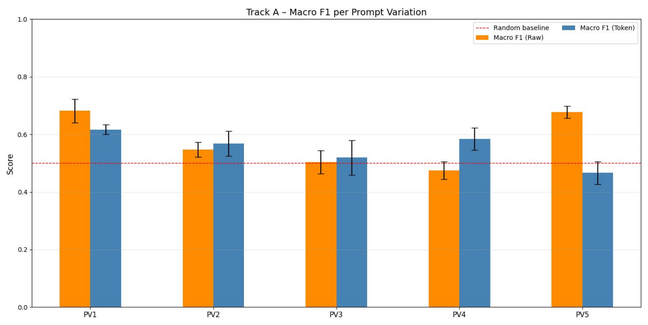
Approach:

- We follow the task baseline and use **GPT-4o-mini** for all experiments.
- Five prompt variations (PV) are tested:
  - **PV1**: Short baseline prompt (task-provided)
  - **PV2**: Structured prompt with explicit definitions and a step-by-step decision procedure
  - **PV3**: Condensed version of PV2 without procedural details
  - **PV4**: PV2 extended with scoring for each dimension; the final choice is based on total points
  - **PV5**: PV1 with an additional line to address narrative similarity dimensions
- Temperature fixed at 0 for (more) deterministic outputs
- Evaluated via five independent runs on raw and tokenized texts per PV
- Performance is measured using **Macro F1-score**

Results:



- The baseline prompt achieves the **best results on raw data**.
- PV5 performs on par with PV1
  → **simple prompts** are **more effective** than complex ones.
- Structured prompts perform better on tokenized data
  → reduced distraction from entity information
- Tokenization lowers performance for the baseline prompt variants
  → Entities provide narrative signals which support similarity judgments in less constrained prompts
- Few-shot learning (20/80 split) boosts structured prompts but not beyond the raw baseline results
- Structured prompts gain additional improvement with tokenized data.



Track B:

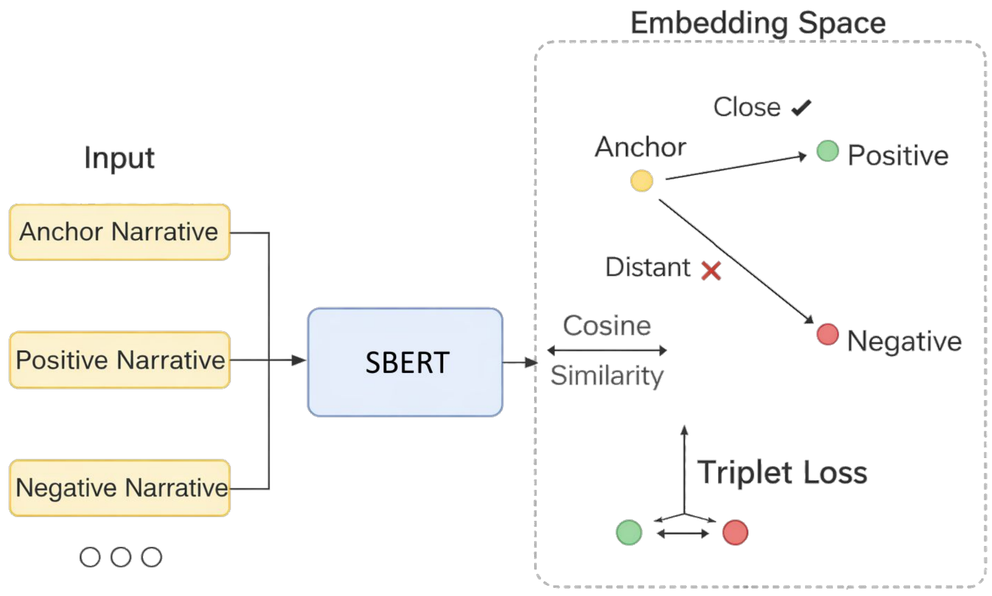
Approach:

Baseline

- Utilizing a higher performance embedding model (**MiniLM** → **MPNet**)
- Applying chunk based embeddings for long sentences.
- Re-evaluating hard cases with tail embeddings.

Fine Tuning

- We used a Sentence-BERT (**SBERT**) model for fine-tuning using triplet loss with cosine similarity
- The **contrastive learning** on the triplets pulls anchor and positive examples closer and pushes negative examples further away.
- Fine-tuning is performed using **MPNet** as the base model.
- These embeddings are evaluated indirectly via triple-wise similarity judgements from Track A.



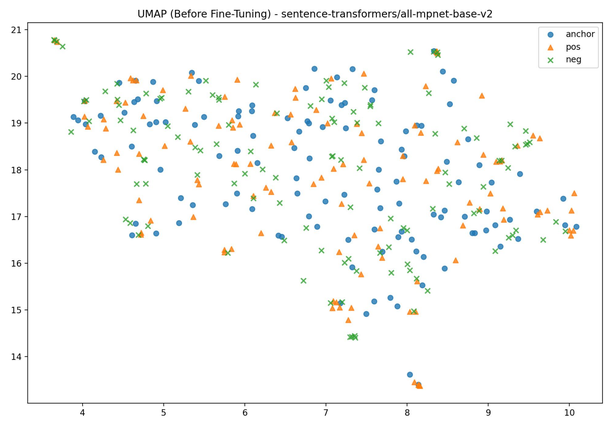
Results:

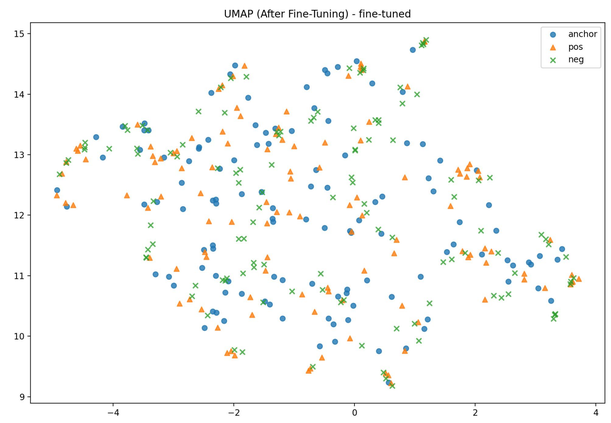
Baseline:

- **MiniM** reached **0.55** accuracy, while **MPNet** improved it to **0.615.**
- Chunk embedding preserved an accuracy of **0.615** for long texts.
- Re-evaluating hard cases further increased Accuracy to **0.625**.

Fine Tuned:

- The fine-tuned model significantly outperforms the non-fine-tuned baseline.
- Best results were achieved without **NER**.
- Hyperparameter tuning resulted in only minor performance variations.
- Improved accuracy and F1-Score to **0.802**.
- Fine-tuning leads to clearer separation of positive and negative examples in embedding space.





Conclusion:

Track A:

- Simple prompts stay superior, even with few-shot learning and especially when narrative entities are preserved
- Complex prompts hinder narrative understanding but improve with few-shot learning on tokenized data

Track B:

- Baseline accuracy was improved to 0.625 through lightweight embedding level enhancements
- Fine-tuning with triplet loss substantially improves the alignment of embeddings with narrative similarity, yielding a clear performance gain

UNIVERSITÄT BONN INFORMATIK

CAISA Lab